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High-Intensity Interval Exercises' Acute Impact on Heart Rate Variability: Comparison Between Whole-Body and Cycle Ergometer Protocols

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Abstract

Schaun, GZ and Del Vecchio, FB. High-intensity interval exercises' acute impact on heart rate variability: comparison between whole-body and cycle ergometer protocols. *J Strength Cond Res* 32(1): 223–229, 2018—Study aimed to compare the effects of 2 high-intensity interval training (HIIT) protocols on heart rate variability. Twelve young adult males (23.3 ± 3.9 years, 177.8 ± 7.4 cm, 76.9 ± 12.9 kg) volunteered to participate. In a randomized cross-over design, subjects performed 2 HIIT protocols, 1 on a cycle ergometer (Tabata protocol [TBT]; eight 20-second bouts at 170% P_{max} interspersed by 10-second rest) and another with whole-body calisthenic exercises (McRae protocol; eight 20-second all-out intervals interspersed by 10-second rest). Heart rate variability outcomes in the time, frequency, and nonlinear domains were assessed on 3 moments: (a) presession; (b) immediately postsession; and (c) 24 hours postsession. Results revealed that RRmean, Ln rMSSD, Ln high frequency (HF), and Ln low frequency (LF) were significantly reduced immediately postsession ($p \leq 0.001$) and returned to baseline 24 h after both protocols. In addition, LF/HF ratio was reduced 24 h postsession ($p \leq 0.01$) and SD2 was significantly lower immediately postsession only in TBT. Our main finding was that responses from heart rate autonomic control were similar in both protocols, despite different modes of exercise performed. Specifically, exercises resulted in a high parasympathetic inhibition immediately after session with subsequent recovery within 1 day. These results suggest that subjects were already recovered the day after and can help coaches to better program training sessions with such protocols.

Key Words interval training, intermittent training, calisthenics, autonomic modulation

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Introduction

High-intensity interval training (HIIT) has been extensively researched and applied to improve sports performance (7,42) as well as to enhance physical fitness (26,37). In this context, comprehension about protocols' acute responses is important to better program and organize training sessions. Previously, outcomes such as heart rate (HR), oxygen consumption ($\dot{V}O_2$) and lactate concentration (16), as well as acute neuromuscular impact (39) and heart rate variability (HRV; 6) were evaluated to better understand demands imposed by HIIT on the body (7,15). Despite knowledge on HIIT influence on several cardiac outcomes and, in particular, on autonomic nervous system (ANS) has been expanded in recent years (29,42), it is still necessary to clarify the effects of exercise mode on HRV.

Heart rate variability is understood as the variation between successive heart beats caused by sympathetic and parasympathetic tones modulation (6). In normal condition, subjects who present greater variability tend to be better conditioned (6). Recent investigations have shown that ANS assessment through HRV is an appropriate strategy to evaluate cardiovascular overload and stress imposed by exercise (2,36). Moreover, HRV indices can also be used to control recovery and parasympathetic reactivation after an exercise session (9,21,36), as well as to program and even prescribe the exercise session itself (2,21,33).

Specifically, regarding postexercise responses, literature seems to agree that the most determinant variable in the magnitude of session's impact on HRV is intensity. When compared with moderate intensity exercise, high intensity tends to present slower HRV recovery because of superior sympathetic activation and increased metabolic by-products accumulation (e.g., H^+, Pi, and lactate; 6). However, there are results that do not corroborate to this difference between training zones (35).

As mentioned earlier, knowing the impact HIIT has on HRV can contribute to comprehend time needed for proper recovery between sessions. Parasympathetic activity return to basal levels can be used as a cardiovascular system

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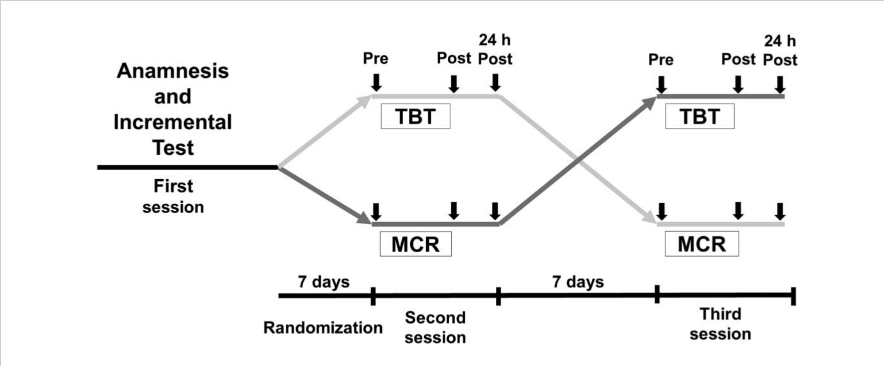

Figure 1. Experimental design of the study. TBT = Tabata protocol; MCR = McRae protocol; arrows indicate heart rate variability measures.

recovery marker to guide new training sessions (21,22,36). Previous studies have demonstrated greater increases in maximal $\dot{V}O_2$ ($\dot{V}O_2max$) when sessions were prescribed based on parasympathetic activity recovery compared with a predefined session schedule (21,22).

Although impact of maximal and supramaximal HIIT on HRV has previously been investigated (8,9), differences between protocols with similar effort:pause relationships but performed with different exercises have not yet been investigated (10). Less attention was also given to training protocols that involved whole-body exercises, such as calisthenics (16,26). Consequently, understanding HRV dynamics in supramaximal and calisthenic HIIT can improve training sessions' prescription. Previously, running was compared with calisthenics and although both achieved HR_{peak} percentages above 90%, calisthenic protocol led to a greater disruption in HRV (24). However, in this investigation, both sessions were held continuously. Thus, intermittent whole-body calisthenic exercise effects on HRV are not yet known.

Therefore, the purpose of this study was to compare cardiac and autonomic responses with 2 HIIT protocols, with same temporality, but performed with different exercise modes. Our initial hypothesis was that both protocols physiological and autonomic responses would not be significantly different.

Methods

Experimental Approach to the Problem

Current investigation had a randomized cross-over designed in which protocol performed was considered as independent variable, whereas HR and HRV outcomes as dependent variables. Participants attended 3 different sessions (Figure 1). During the first, anthropometric data were collected and an incremental test to measure power associated with maximal oxygen consumption (P_{max}) was performed on a cycle ergometer (BIOTEC 2100; CEFISE, Belo Horizonte, Brazil). Incremental test was preceded by a 10-minute warm-up at ~90 rpm and around 60% estimated HR_{max} (32). After resting 5 minutes, test started with a 0.6-Kp load and, every min, load was increased by 0.25 Kp until volitional exhaustion. Final load corresponding to a completed stage was considered as P_{max}.

One week after first session, participants were randomized between 2 experimental protocols and performed the protocol to which they were assigned. Seven days apart from the second session, subjects performed remaining protocol. All tests were performed in the morning, and laboratory temperature was kept between 21 and 24° C. In addition, subjects were advised not to consume stimulants such as caffeine or alcohol and not to perform vigorous physical activities 24 hours before the tests. They were also asked to maintain their normal routine (e.g., diet and sleep) and to avoid vigorous exercise during the week between experimental sessions.

Subjects

Twelve physically active young adult males (Mean +/− SD 23.3 ± 3.9 years, 177.8 ± 7.4 cm, 76.9 ± 12.9 kg) volunteered

Table 1. Cardiorespiratory responses according to protocol performed in young men ($n = 12$).*†

	TBT			MCR		
	Pre	Exercise	24 h post	Pre	Exercise	24 h post
HR_{avg} ($b \cdot min^{-1}$)	65.4 ± 6.3	144.2‡ ± 11.9	66.0 ± 6.0	63.1 ± 6.7	130.6‡ ± 10.4	65.4 ± 6.0
HR_{peak} ($b \cdot min^{-1}$)	79.8 ± 9.8	176.6‡ ± 9.8	81.5 ± 6.5	80.0 ± 8.8	169.7‡ ± 9.4	82.5 ± 6.5
$\%HR_{peak}$	–	87.9 ± 4.2	–	–	86.9 ± 5.5	–

*TBT = Tabata protocol; MCR = McRae protocol; Pre = presession; Exercise = during protocol; 24 h post = 24 hours postsession.
†Values are mean ± *SD*.
‡Significantly higher than pre and 24 hours post ($p \leq 0.001$).

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to participate in the study. Those who presented lower limb injuries, muscle, and/or join pain in the past 3 months, cardiometabolic problems or regular medication usage were not included. Before any procedure, volunteers read, agreed, and signed an informed consent, and research project had already been approved by the School of Physical Education at Federal University of Pelotas' ethics committee (004/2012) in accordance with the Declaration of Helsinki.

Sample size was calculated a priori, based on a previous study comparing HRV responses in continuous calisthenic and running sessions (24). These sessions resulted in Ln rMSSD values equal to 1.92 ± 0.35 ms^2 and 2.76 ± 0.55 ms^2. Considering an 80% power and 5% significance level, a sample of 10 subjects would be necessary. An additional 20% was added to account for possible losses or refusals, resulting in a final sample size of 12 men.

Procedures

High-Intensity Interval Exercise Protocols. One protocol was performed on a cycle ergometer (Tabata protocol [TBT]) as described by Tabata et al. (37). Briefly, it was composed by 5-minute warm-up at around 90 rpm and 60% estimated HR$_{max}$ followed by 5-minute rest. Then, subjects executed eight 20-second bouts at 170% P_{max} and above 85 rpm interspersed by 10-second passive recovery. Tabata protocol was performed on the same ergometer as the incremental test in conjunction with Ergometric 6.0 software (CEFISE) to proper control cadence and time elapsed.

Calisthenic protocol (McRae protocol [MCR]) was adapted from McRae et al. (26). Subjects were submitted to 2 repetitions of a predefined sequence composed by 4 whole-body exercises (burpees, mountain climbers, jumping jacks, and squat and thrusts with 3.1 kg dumbells). In total, eight 20-second efforts interspersed by 10-second passive rest were performed at an all-out intensity. It should be noted that during both experimental protocols, subjects received strong verbal stimulation to ensure that they performed maximal effort possible.

Heart Rate Variability. All variables related to HRV were assessed in 3 moments: (a) immediately presession; (b) immediately postsession; and (c) 24 hours postsession. In all 3 moments, continuous R-R intervals were acquired throughout 10 minutes in supine position. For such, subjects wore a HR monitor (RS800CX; POLAR, Kempele, Finland), which was previously validated for HRV analysis (18). Data were downloaded to computer through Polar Pro-Trainer 5.0 software, and HRV analysis was performed using Kubios HRV software (Kuopio, Finland). First, R-R intervals were visually inspected and adjacent beats were interpolated when necessary. Then, a 5-minute period free of noise was selected for analysis and all HRV variables were calculated (13). It should be noted that this time frame (i.e., 5 minutes) seems sufficient to assess and analyze data related to HRV (12). Specifically, the mean between successive R-R intervals (RRmean), natural logarithm of the root-mean-square of

Table 2. Heart rate variability variables in the time and frequency domains according to protocol performed in young men (n = 12).*†

	TBT			MCR				
	Pre	Post	24 h post	Pre	Post	24 h post	ES	ES$_{24}$
Time domain								
RRmean (ms)	910.1 ± 89.7	450.1 ± 92.1‡	919.4 ± 91.0	961.7 ± 96.6	476.0 ± 41.3‡	956.6 ± 102.0	0.84	0.30
Ln rMSSD (ms)	3.82 ± 0.27	1.98 ± 0.45‡	3.65 ± 0.49	3.82 ± 0.27	1.76 ± 0.21‡	3.73 ± 0.42	0.30	0.16
Frequency domain								
Ln LF (ms^2)	7.68 ± 0.56	3.41 ± 1.48‡	7.03 ± 0.60	7.50 ± 0.45	3.31 ± 0.65‡	7.41 ± 0.57	0.26	0.71
Ln HF (ms^2)	6.51 ± 0.57	1.44 ± 0.79‡	6.13 ± 1.21	6.60 ± 0.59	1.99 ± 0.87‡	6.54 ± 1.17	0.52	0.34
LF/HF	3.61 ± 1.94	5.44 ± 1.85	2.81 ± 2.51§	2.75 ± 1.54	5.11 ± 3.88	1.74 ± 0.91§	0.75	0.60

*TBT = Tabata protocol; MCR = McRae protocol; Pre = presession; Post = immediately postsession; 24 h post = 24 hours postsession; ES = immediately postsession between-groups' effect size; ES$_{24}$ = 24 hours postsession between-groups' effect size; LF = low frequency; HF = high frequency.
†Values are mean ± SD.
‡Significantly different from pre and 24 hours post ($p \le 0.001$).
§Significantly different from post ($p \le 0.01$).

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successive differences between adjacent R-R intervals (Ln rMSSD), SD1 and SD2, natural logarithm of the high (Ln HF; 0.15–0.40 Hz) and low (Ln LF; 0.04–0.15 Hz) frequency components, as well as the LF/HF ratio were assessed.

Generally, HRV investigations control respiratory rate, however, to maintain HR natural return we chose not to perform such control, thus, increasing results' external validity and applicability to practice, as suggested previously (2,8). Average and peak HR (HR_{avg} and HR_{peak}, respectively) were assessed during presession and 24 hours postsessions together with HRV and also during MCR and TBT protocols. In addition, based on HR_{avg} values during incremental test, HR_{avg} percentage in relation to HR_{peak} (% HR_{peak}) was calculated.

Statistical Analyses

Data normal distribution was tested with Shapiro-Wilk test, and results are presented as mean and $\pm SD$. To analyze % HR_{peak}, a t test for dependent samples was used. For the other variables, repeated-measures 2-way analysis of variances (protocol $\times$ moment) were applied and $p \leq 0.05$ was considered as statistically significant. Finally, between-group Cohen's d effect sizes (ES) were calculated for the immediately and 24 hours postsession moments as follows: "(group 1 mean change $-$ group 2 mean change)/pooled SD of the change score," as previously proposed (11). Specifically, ES values were considered as trivial (<0.35), small (0.35–0.80), moderate (0.80–1.50), and large (>1.5) in accordance with Rhea (31).

Results

Heart rate comparisons are presented in Table 1. HR_{avg} and HR_{peak} were higher during session in comparison with presession and 24 hours postsession moments ($p \leq 0.001$ for both) with no difference between protocols. In addition, % HRpeak immediately postsession corresponded to 87.9 $\pm$ 4.15% and 86.9 $\pm$ 5.46% during TBT and MCR, respectively, with no differences between them ($p = 0.91$).

In relation to ANS modulation, both protocols presented similar behaviors (Table 2). All variables were significantly reduced immediately postsession ($p \leq 0.001$) in comparison with presession. Also, HRV measures returned to basal levels 24 hours after both protocols ($p \leq 0.001$).

Figure 2 presents nonlinear HRV results. In both protocols, SD1 behaved similarly, with a significant reduction after the sessions ($p \leq 0.01$; ES = 0.46) which also returned to baseline 24 hours after (ES = 0.84). Regarding SD2, both TBT and MCR showed a significant reduction immediately after session ($p \leq 0.01$; ES = 5.45) which persisted 24 hours post only in TBT group ($p \leq 0.05$; ES = 4.0), whereas MCR had already returned to baseline the day after.

Discussion

This study aimed to investigate and compare the effects of 2 HIIT protocols on HR and its variability. Our main finding was that HR autonomic control behaved similarly in both protocols, despite different modes of exercise performed. Specifically, exercises resulted in a high parasympathetic inhibition immediately after session with subsequent recovery within 1 day.

All variables analyzed in time and frequency domains (i.e., RRmean, Ln rMSSD, Ln HR, and Ln LF) showed immediate reductions. Such already expected behavior occurred because of vagal inhibition in addition to an increase in sympathetic stimulation in response to the high-intensity used (1,14,19). However, this result had not yet been analyzed in whole-body calisthenic exercises. Previously, high-intensity whole-body exercises performed in a continuous manner were shown to generate higher HRV disruption when compared with high-intensity continuous running (24). This, way, absence of differences between protocols should also be emphasized, as this is the first study to compare autonomic regulation in HIIT whole-body and ergometer protocols.

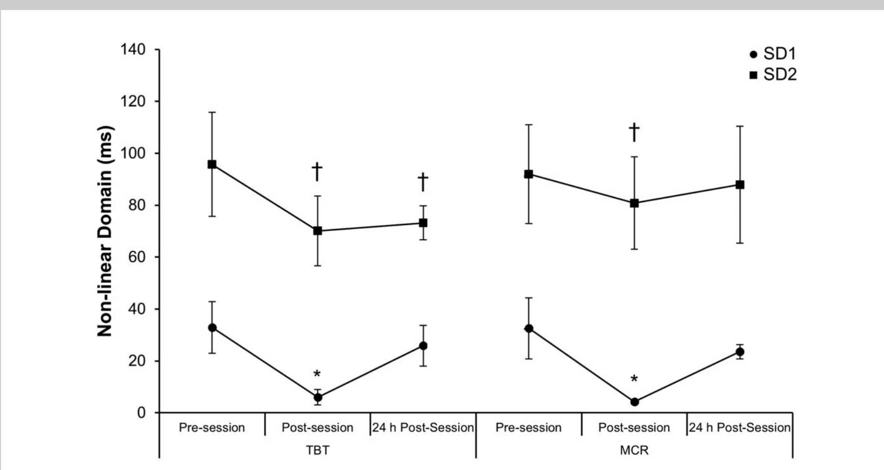

Figure 2. Heart rate variability nonlinear domain analysis according to protocol ($n = 12$); TBT = Tabata protocol; MCR = McRae protocol; * = significantly different from presession and 24 hours postsession ($p \leq 0.01$); † = significantly different from presession ($p \leq 0.05$).

Because of the very high intensities used in both protocols, it was expected that autonomic cardiac stress would last for a longer period. However, on the contrary, analyzed variables returned to resting levels on the following day, as previously demonstrated in

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supramaximal protocols (1,27). This fact seems to indicate, from a cardiovascular stress perspective, that subjects were recovered and able to perform a new training session (29,42). From this same premise, previous investigations have already observed an improvement in aerobic fitness when prescribing aerobic training sessions based on HRV return to baseline (21,22), highlighting the relevance our results can have to those planning exercise sessions distribution over a given training period.

Despite this, attention should be paid to other systems involved in the exercise because neuromuscular (39) and/or metabolic (20) components may not be fully recovered in the same time interval after a HIIT session. Therefore, taking into account other activities athletes need to perform during a week, for example, is of paramount importance for coaches. Repetition of protocols without proper recovery of these systems may impair other sessions, such as technical ones, or even prevent adequate intensity maintenance in subsequent HIIT sessions (7,28). Consequently, this could result in suboptimal adaptations arising from training cycles in medium or long term.

Still, an exception was observed in LF/HF ratio, which did not present reduction immediately after TBT or MCR but, instead, was reduced 24 hours postsession. Traditionally, this variable is used as an indicator of sympathovagal modulation (5). Increase in the ratio would indicate sympathetic modulation predominance, whereas reduction would suggest parasympathetic branch predominance. In this perspective, reduction 24 hours later in LF/HF would point to a "supercompensation" in vagal modulation (23,36). Some authors suggest that this situation or period would be ideal to apply new training sessions, in line with classic sport training theories. However, although inferring supercompensation based on this result is tempting, caution is needed. Although widely used in exercise sciences, recent studies have questioned LF/HF ratio reliability (4,5). Previously, it was believed that parasympathetic branch would be responsible for HF and sympathetic for LF. Today, it is estimated that approximately 90 and 50% of observed variations in HF and LF are due to the parasympathetic branch, whereas sympathetic one would influence 10 and 25% of same variables, respectively. Yet, the remaining 25% LF variation would not be explained by parasympathetic or sympathetic components of ANS (for a review, see Ref. 5). Because of this complex interaction, presuming that LF/HF reduction observed in the current study is directly related to parasympathetic predominance may be too simplistic. Nevertheless, regardless of vagal supercompensation or not, other variables analyzed point to a parasympathetic return to basal levels 24 hours postsession. Such a condition would already be adequate to allow new sessions to be performed.

In nonlinear HRV domain, SD1 represents an instantaneous index of beat-to-beat variability (38,40,41). As with the other variables, SD1 also showed reduction after exercise, which may have been motivated by decreased variability between R-R intervals (38). Its reduction points to an overloaded cardiac system performance because of the high demand imposed by the protocols. Furthermore, SD2, which represents oscillations in long-term variability (38) was shown to be reduced after exercise and, interestingly, remained reduced 24 hours after TBT. Previous study observed reduction in SD2 after complete parasympathetic blockade in maximum incremental exercise (38). According to authors, this reduction would be associated with an increase in sympathetic activation. Thus, sympathetic activation may have remained elevated 24 hours after TBT protocol, despite vagal modulation restoration, as evidenced from Ln HF, for example (6,12). Such difference between protocols (i.e., TBT vs. MCR) could be associated with how they were performed. That is, TBT primarily involved the same musculature throughout all series, whereas MCR alternated movement pattern between effort intervals. Increased recruitment of the same musculature may have led to greater anaerobic engagement and metabolites accumulation possible leading to changes in sympathetic stimulation for a longer period (1). An acute increase in plasma volume may have also interfered with these results, although less likely because of protocol volume used (9). Previously, it was demonstrated that SD2 is able to provide information about HR dynamics that would not be possible through linear variables, demonstrating the latter would be poor indicators to track changes in sympathetic tone. Our results corroborate this notion because no differences were found in time and frequency domain outcomes but only in SD2. However, this assumption needs to be better tested before definitive conclusions can be made. Furthermore, based on these results it could also be suggested that HIIT protocols that require a variety of movements may be advantageous compared with "monotonous" cycling movements in physically active men to enhance recovery.

Regarding HR_{avg} and HR_{peak}, there were no differences between protocols. In addition, stimuli in both training modes corresponded to ~87% HR_{peak} showing their intensity is equivalent and can be defined as vigorous according to the American College of Sports Medicine (17). In this context, sprint interval training (16) and calisthenic continuous training (24) were also shown to elicit %HR_{peak} close to 90%. Ergometer-based HIIT has been widely accepted as an efficient method to improve cardiovascular fitness in athletes (7,25) and adults (34,43) for half a century (3,7). Meanwhile, HIIT based on whole-body exercises was only shown as effective recently (26). Thus, longitudinal studies evaluating adaptive responses to these protocols are warranted to further comprehend their potentials in the long term. Also, considering subjects were already recovered 24 hours after sessions, it is suggested to evaluate whether performing 2 or more sessions with a 24-hour interval between them will allow adequate HRV recovery. As limitations, although subjects were asked to refrain from vigorous exercise and maintain their normal routine during the week between

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experimental sessions, this was not directly assessed in the presented study. Furthermore, we were unable to perform collections in less than 24 hours postsession, so our finding may not be extrapolated for shorter periods and it is still necessary to determine precisely when subjects are already recovered. This information may be important for subjects who have less than 24 hours between training sessions or perform more than 1 session per day. In conclusion, both HIIT protocols affected ANS similarly. Furthermore, despite the vigorous intensity observed, subjects' HRV were already recovered 24 hours after the session.

Practical Applications

Trainers and coaches can rely on HRV values to prescribe new training session. Based on our results, 24 hours seems like an appropriate time frame for new sessions to be performed. However, professionals should be aware that other systems (e.g., neuromuscular and/or metabolic) might not be recovered as quickly as the cardiovascular system and may depend on the subject's training status and type of exercise performed. In addition, both protocols presented similar high cardiovascular demands, suggesting that athletes may benefit from using both these protocols to enhance cardiorespiratory fitness and reduce monotony. Moreover, MCR is composed by calisthenics exercises of low complexity and at the maximum intensity with which the subject manages to perform them. Because there is no need for a specific cycle ergometer and previous maximum incremental test for its proper usage, practitioners and coaches can rely on it when those (i.e., ergometer and incremental tests) are not available. As a result, MCR may be an alternative and practical strategy to TBT to stress the cardiovascular system. Furthermore, new methods have been proposed to monitor HRV. As an example, smartphone app HRV4Training was recently shown as both accurate and valid to measure HRV compared with HR monitor and electrocardiography (30). Such innovations may facilitate HRV assessment in the field on a day-today basis and enhance control on athlete's physiological state by coaches.

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